

Probabilistic sensor fusion

Lecture 5: Gaussian processes

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SC42170, EE4760

Q3 2025/2026

Practicalities

- Final assessment is on Wednesday April 8th 11:00 - 12:00.
- Next week Wednesday there is no lecture (but there is an exercise session on Friday and next week Friday!)

Previous lectures

State space model:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{w}_k)$$

$$\mathbf{y}_k = h(\mathbf{x}_k, \mathbf{v}_k)$$

Goal: Approximate the posterior $p(\mathbf{x}_k \mid \mathbf{y}_{1:k})$.

Previous lectures

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Today: Assume that part of the model is unknown and learn it using Gaussian processes.

Remember from Filtering & Identification

- **Filtering:** Estimate the state of a time-varying state space model:

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k & \text{or} & & \mathbf{x}_{k+1} &= f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k & & \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \\ \mathbf{y}_k &= \mathbf{C}\mathbf{x}_k + \mathbf{D}\mathbf{u}_k + \mathbf{v}_k & \text{or} & & \mathbf{y}_k &= h(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{v}_k & & \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})\end{aligned}$$

given the models $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ and input-output data $\mathbf{u}_{1:k}, \mathbf{y}_{1:k}$.

- **Identification:** Identify parametric model functions $f(\boldsymbol{\theta})$ and $h(\boldsymbol{\theta})$ or matrices $(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D})$ from input-output data $\mathbf{u}_{1:K}, \mathbf{y}_{1:K}$.

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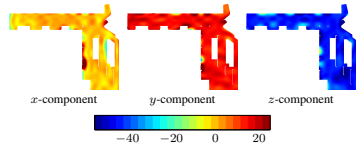
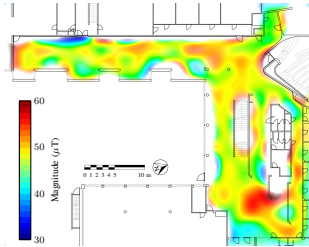
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This lecture: Learn a **non-parametric** model instead.

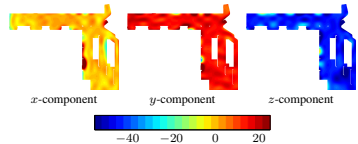
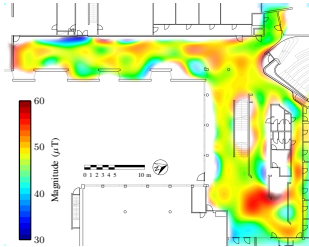
Example: Map of the magnetic field

Difficult to model in terms of a number of parameters $\Rightarrow$ Learn a continuous magnetic field map.



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Uncertainty of this map is important!

Gaussian processes

- **Multivariate Gaussian distributions**
- **Gaussian process priors**
- Gaussian process regression

Remember

- ① A Gaussian distribution is fully described by its mean and covariance.

$$\mathbf{x} \sim \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_x, \mathbf{P}_x)$$

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- 3 A posterior can be computed using Bayes' theorem

Bayes theorem:
$$p(\mathbf{a} \mid \mathbf{b}) = \frac{p(\mathbf{b} \mid \mathbf{a})p(\mathbf{a})}{p(\mathbf{b})}.$$

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- 4 Assume that the random variables $\mathbf{x}_a$ and $\mathbf{x}_b$ are jointly distributed as

$$p(\mathbf{x}_a, \mathbf{x}_b) = \mathcal{N} \left(\begin{pmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{pmatrix}; \begin{pmatrix} \boldsymbol{\mu}_a \\ \boldsymbol{\mu}_b \end{pmatrix}, \begin{pmatrix} \mathbf{P}_{aa} & \mathbf{P}_{ab} \\ \mathbf{P}_{ba} & \mathbf{P}_{bb} \end{pmatrix} \right).$$

Then the marginal densities $p(\mathbf{x}_a)$ and $p(\mathbf{x}_b)$ are given by

$$p(\mathbf{x}_a) = \mathcal{N}(\mathbf{x}_a; \boldsymbol{\mu}_a, \mathbf{P}_{aa}), \quad p(\mathbf{x}_b) = \mathcal{N}(\mathbf{x}_b; \boldsymbol{\mu}_b, \mathbf{P}_{bb}).$$

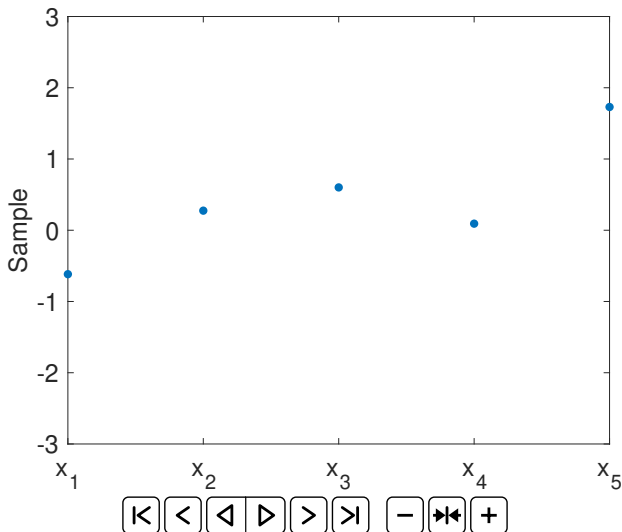
Sampling from an independent multivariate Gaussian

Sample from:

$$\begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \right)$$

(using `mvnrnd`).

Sampling from an independent multivariate Gaussian



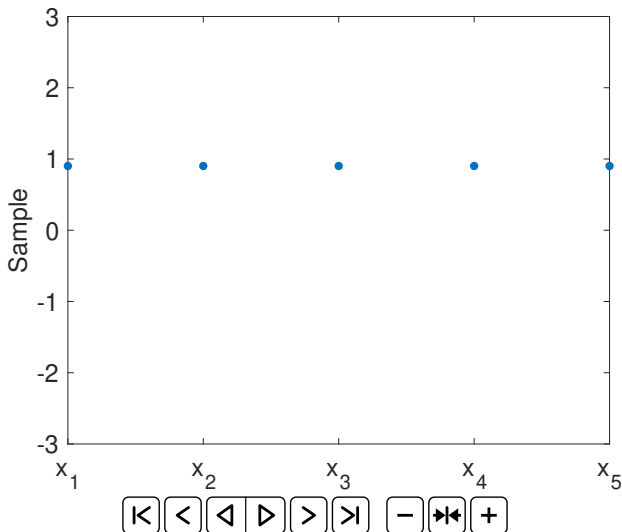
Sampling from a fully correlated multivariate Gaussian

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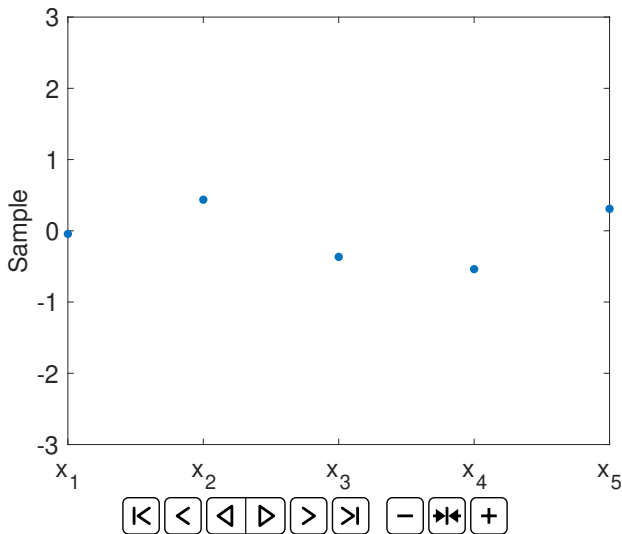
Sampling from a partially correlated multivariate Gaussian

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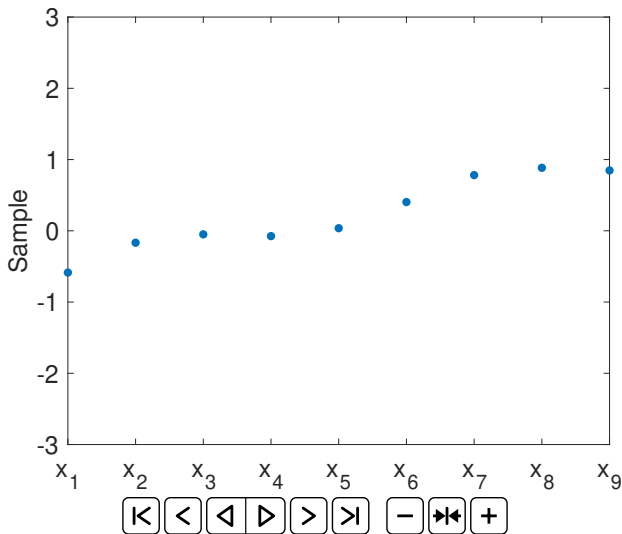
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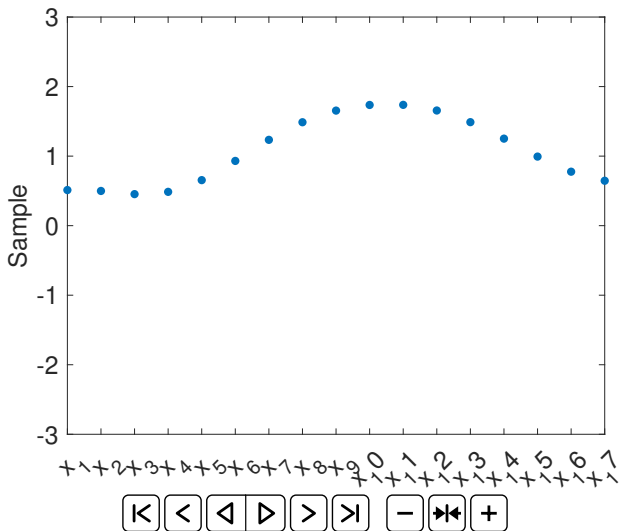
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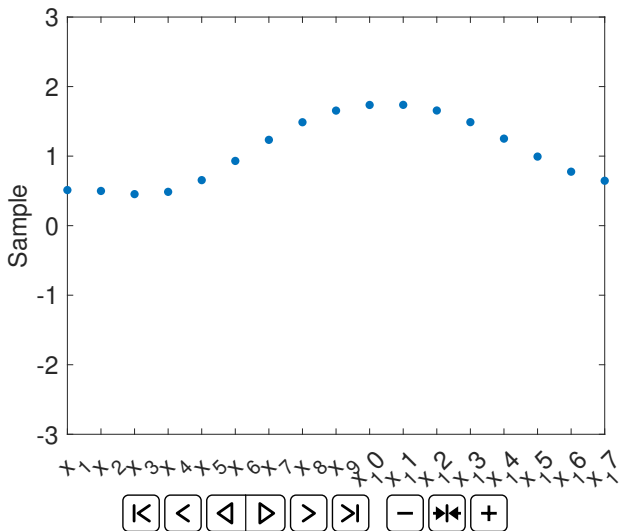
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This looks like a function!! Can we use this for modelling?

Gaussian process priors (1)

- Distribution over functions $f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}'))$.
- The prior mean function is often chosen to be zero.
- Prior covariance function (kernel) encodes properties like smoothness, amount and lengthscale of variations.
- There are many choices for prior covariance functions, the most common one is the squared exponential

$$\kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right)$$

Gaussian process priors (2)

If $f(\mathbf{x}) \sim \mathcal{GP}(0, \kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}'))$ with $\kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\frac{\|\mathbf{x}-\mathbf{x}'\|_2^2}{2\ell^2})$, then

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Remember the Gaussian property related to marginalization!

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When

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the marginalization property implies that

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Gaussian process priors (4)

Going back to our example of sampling from a correlated Gaussian distribution

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- I chose to plot $f_1, \dots, f_5$ at equidistant locations $x_1, \dots, x_5$ in ascending order on the x-axis.

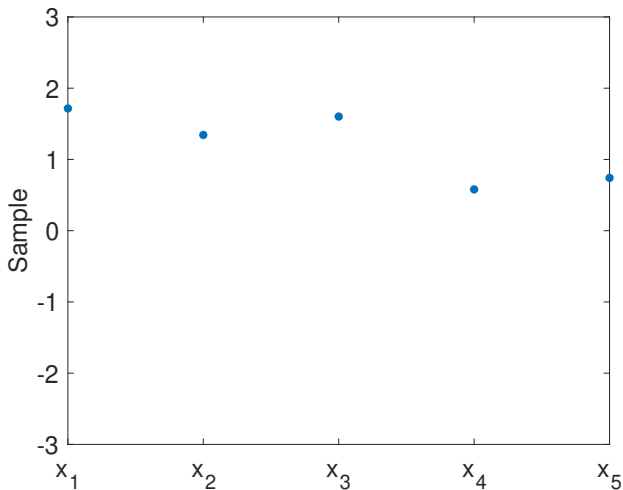
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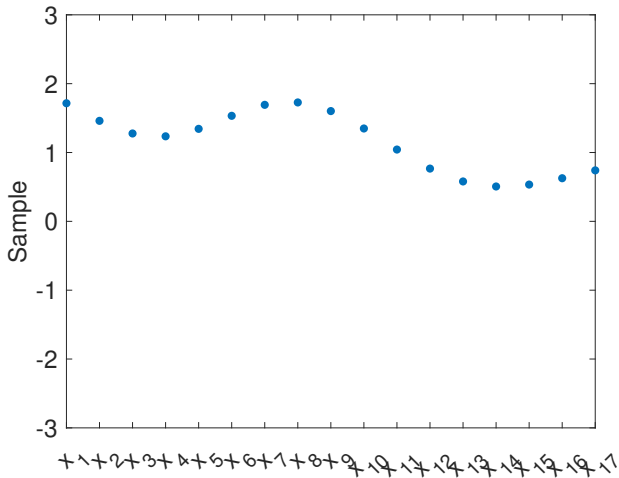
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- I chose to plot $f_1, \dots, f_5$ at equidistant locations $x_1, \dots, x_5$ in ascending order on the x-axis.
- I used a squared exponential kernel with $\sigma_f = 1$, $\ell = 1$. Note that $\exp(-\frac{(0)^2}{2}) = 1$, $\exp(-\frac{(1)^2}{2}) \approx 0.61$, $\exp(-\frac{(2)^2}{2}) \approx 0.14$, $\exp(-\frac{(3)^2}{2}) \approx 0.01$, $\exp(-\frac{(4)^2}{2}) \approx 0$.

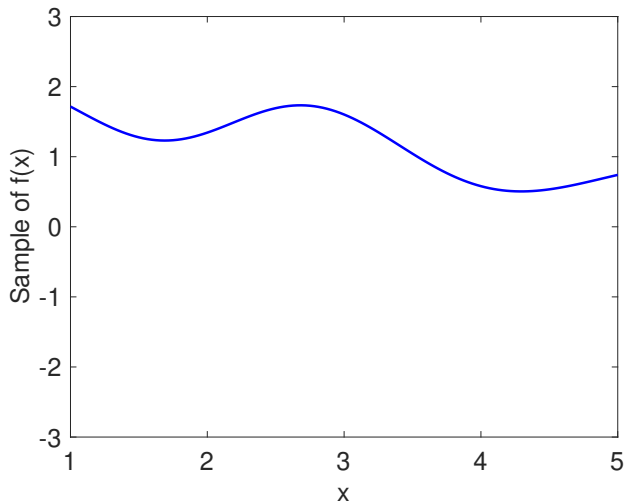
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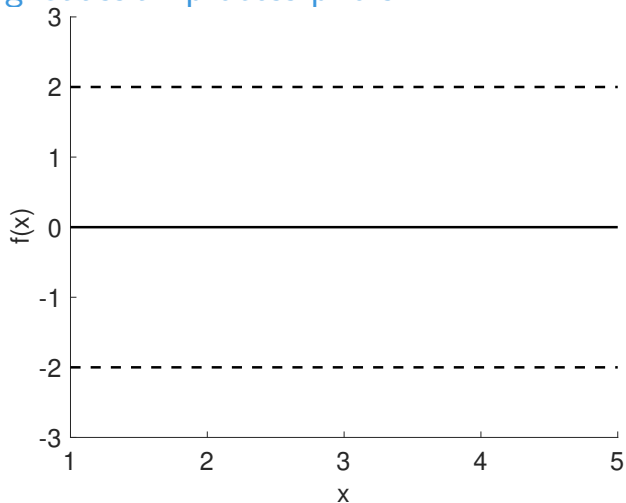
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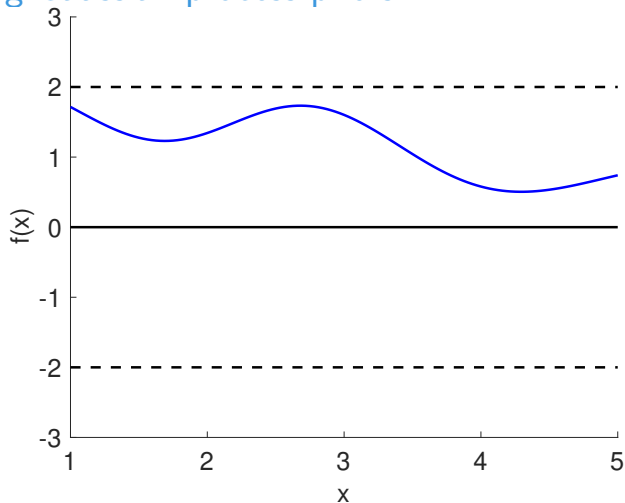
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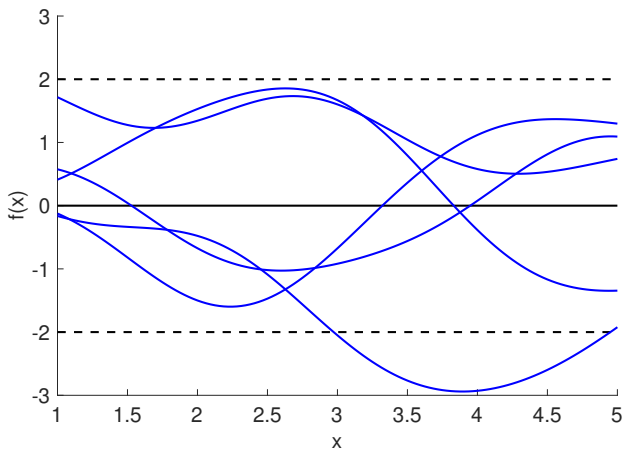
Visualizing Gaussian process priors



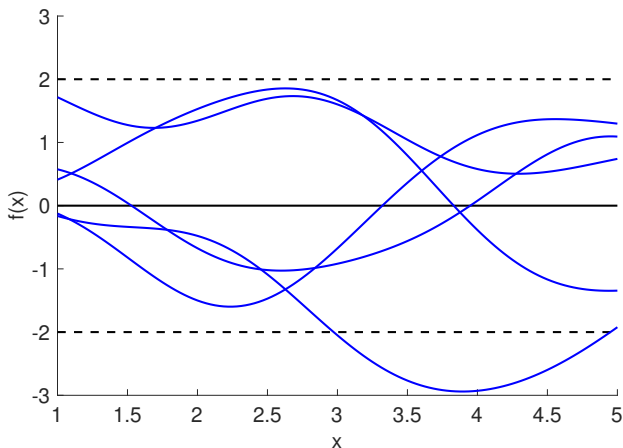
Visualizing Gaussian process priors



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Visualizing Gaussian process priors



In summary, Gaussian process priors can be used to encode prior knowledge about functions, e.g. their smoothness, amount and lengthscale of variations.

Gaussian processes

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Gaussian process regression

Assume that we want to learn a function $f(\mathbf{x})$ based on the prior information $f(\mathbf{x}) \sim \mathcal{GP}(\boldsymbol{\mu}(\mathbf{x}), \kappa(\mathbf{x}, \mathbf{x}'))$ and a measurement of this function at location $\mathbf{x}$, with Gaussian noise

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We can use Bayes rule to compute the posterior of the function f !

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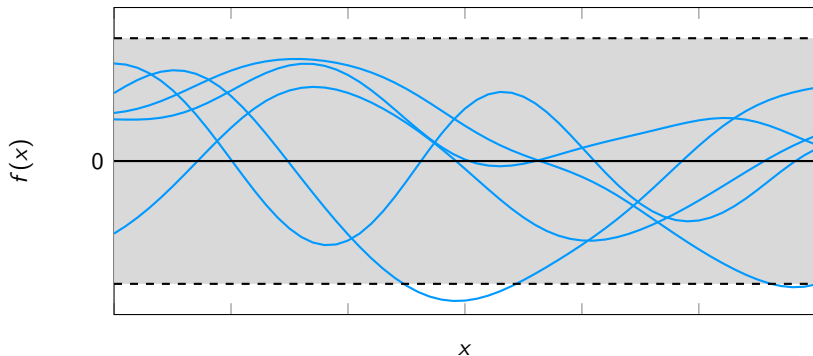
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Since we have a Gaussian process prior and Gaussian measurement noise, the posterior is also a Gaussian process! This implies that:

- We can compute the posterior again in terms of a mean and a covariance.
- We can visualize the posterior Gaussian process in a similar way to before.

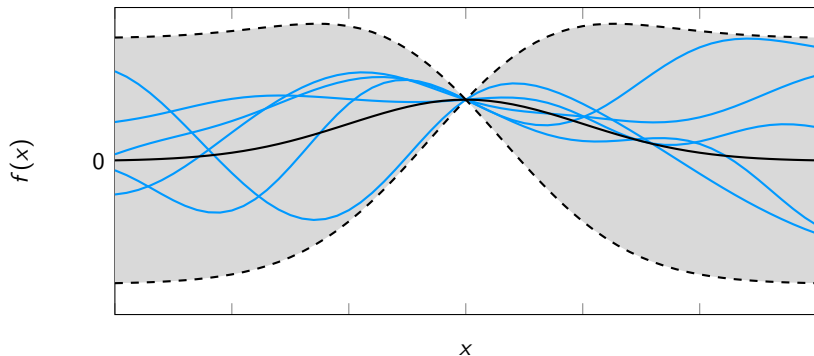
Gaussian process regression visualized

Estimate $f(\mathbf{x})$ using a squared exponential, zero-mean prior and noisy observation(s) $y = f(\mathbf{x}) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$, $\sigma_\epsilon = 0$.



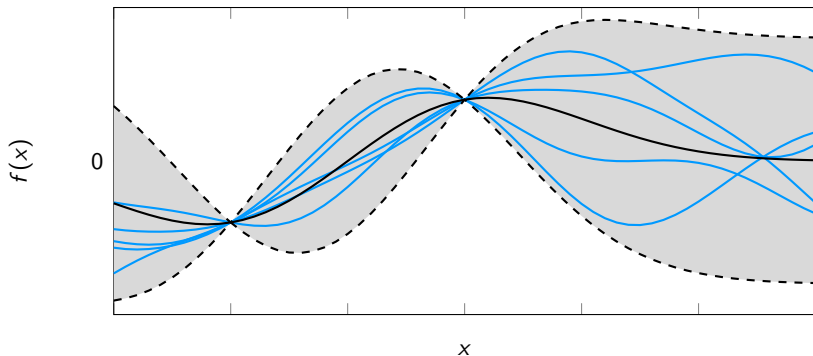
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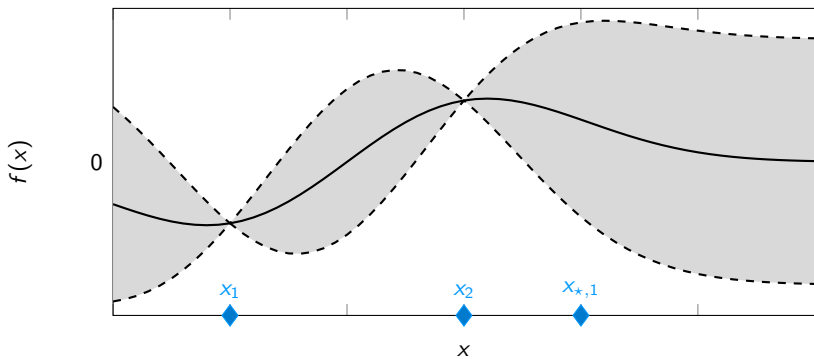


Computing the GP posterior / prediction (1)

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- Instead of deriving the posterior of the entire function $f(\mathbf{x})$, let us compute the posterior of f at some location(s) $\mathbf{X}_\star$.
- In other words, we want to combine the GP prior with (noisy) measurements $\mathbf{y}$ at locations $\mathbf{X}$ to predict the function value(s) $f_\star/\mathbf{f}_\star$ and its uncertainty at location(s) $\mathbf{X}_\star$.



$$\mathbf{X} = \{x_1, x_2\}, \mathbf{X}_* = x_{*,1}$$

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- Let us assume a zero-mean Gaussian process.

Computing the GP posterior / prediction (2)

To compute the $p(\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star)$, let us

- 1 First compute the joint distribution $p(\mathbf{y}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$.
- 2 Then use conditioning to the distribution $p(\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star)$.

Computing the joint distribution $p(\mathbf{y}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$

To compute the joint distribution $p(\mathbf{y}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$

- 1 First, remember slide 11 and 13.

Gaussian process priors (2)

If $f(\mathbf{x}) \sim \mathcal{GP}(0, \kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}'))$ with $\kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\frac{\|\mathbf{x}-\mathbf{x}'\|_2^2}{2\ell^2})$, then

$$\begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \\ \vdots \\ f(\mathbf{x}_\infty) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_f^2 & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1-\mathbf{x}_2\|_2^2}{2\ell^2}) & \cdots & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1-\mathbf{x}_\infty\|_2^2}{2\ell^2}) \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2-\mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 & \cdots & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2-\mathbf{x}_\infty\|_2^2}{2\ell^2}) \\ \vdots & \ddots & \ddots & \ddots \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_\infty-\mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_\infty-\mathbf{x}_2\|_2^2}{2\ell^2}) & \cdots & \sigma_f^2 \end{pmatrix} \right)$$

Gaussian process priors (3)

When

$$\begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \\ \vdots \\ f(\mathbf{x}_\infty) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_f^2 & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2}{2\ell^2}) & \dots & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1 - \mathbf{x}_\infty\|_2^2}{2\ell^2}) \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2 - \mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 & \dots & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2 - \mathbf{x}_\infty\|_2^2}{2\ell^2}) \\ \vdots & \ddots & \ddots & \ddots \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_\infty - \mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_\infty - \mathbf{x}_2\|_2^2}{2\ell^2}) & \dots & \sigma_f^2 \end{pmatrix} \right),$$

the marginalization property implies that

$$\begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_f^2 & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2}{2\ell^2}) \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2 - \mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 \end{pmatrix} \right)$$

Computing the joint distribution $p(\mathbf{y}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$

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- 1 First, remember slide 11 and 13.

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- 1 First, remember slide 11 and 13.
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Some notation related to covariance functions and matrices

Let us assume a Gaussian process prior $f(\mathbf{x}) \sim \mathcal{GP}(0, \kappa(\mathbf{x}, \mathbf{x}'))$.

- For any input locations $\mathbf{X}$ and $\mathbf{X}_*$, we can use the covariance *function* $\kappa(\mathbf{x}, \mathbf{x}')$ to evaluate covariance *matrices* $\mathbf{K}(\mathbf{X}, \mathbf{X})$, $\mathbf{K}(\mathbf{X}_*, \mathbf{X}_*)$ and $\mathbf{K}(\mathbf{X}_*, \mathbf{X})$.

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- For example, if $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2\}$ and our kernel is a squared exponential kernel, then

$$\begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \underbrace{\begin{pmatrix} \sigma_f^2 & \sigma_f^2 \exp(-\frac{\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2}{2\ell^2}) \\ \sigma_f^2 \exp(-\frac{\|\mathbf{x}_2 - \mathbf{x}_1\|_2^2}{2\ell^2}) & \sigma_f^2 \end{pmatrix}}_{\mathbf{K}(\mathbf{X}, \mathbf{X})} \right)$$

Computing the joint distribution $p(\mathbf{y}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$

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- 1 First, remember slide 11 and 13.
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- 3 The joint distribution $p(\mathbf{f}, \mathbf{f}_\star \mid \mathbf{X}, \mathbf{X}_\star)$ is then

$$\mathcal{N} \left(\begin{pmatrix} \mathbf{f} \\ \mathbf{f}_\star \end{pmatrix} ; \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{K}(\mathbf{X}, \mathbf{X}) & \mathbf{K}(\mathbf{X}, \mathbf{X}_\star) \\ \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) & \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) \end{pmatrix} \right)$$

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- 4 Using the measurement model $\mathbf{y} = f(\mathbf{x}) + \epsilon$, $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma_\epsilon^2 \mathbf{I})$, the joint distribution $p(\mathbf{y}, \mathbf{f}_\star | \mathbf{X}, \mathbf{X}_\star)$ is

$$\mathcal{N} \left(\begin{pmatrix} \mathbf{y} \\ \mathbf{f}_\star \end{pmatrix} ; \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I} & \mathbf{K}(\mathbf{X}, \mathbf{X}_\star) \\ \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) & \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) \end{pmatrix} \right)$$

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Conditional distribution of two Gaussian variables

If the joint distribution of two random variables $\mathbf{x}, \mathbf{y}$ is given by

$$p(\mathbf{x}, \mathbf{y}) = \mathcal{N} \left(\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} ; \begin{pmatrix} \boldsymbol{\mu}_x \\ \mathbf{H}\boldsymbol{\mu}_x \end{pmatrix}, \begin{pmatrix} \mathbf{P}_x & \mathbf{P}_x\mathbf{H}^\top \\ \mathbf{H}\mathbf{P}_x & \mathbf{H}\mathbf{P}_x\mathbf{H}^\top + \mathbf{R} \end{pmatrix} \right)$$

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then the conditional distribution of $\mathbf{x}$ given $\mathbf{y}$ is given by

$$p(\mathbf{x} | \mathbf{y}) = \mathcal{N} \left(\mathbf{x}; \boldsymbol{\mu}_x + \mathbf{P}_x\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_x\mathbf{H}^\top + \mathbf{R} \right)^{-1} (\mathbf{y} - \mathbf{H}\boldsymbol{\mu}_x), \right. \\ \left. \mathbf{P}_x - \mathbf{P}_x\mathbf{H}^\top \left(\mathbf{H}\mathbf{P}_x\mathbf{H}^\top + \mathbf{R} \right)^{-1} \mathbf{H}\mathbf{P}_x \right)$$

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If the joint distribution of two random variables $\mathbf{x}, \mathbf{y}$ is given by

$$p(\mathbf{f}_*, \mathbf{y} \mid \mathbf{X}, \mathbf{X}_*) = \mathcal{N} \left(\begin{pmatrix} \mathbf{f}_* \\ \mathbf{y} \end{pmatrix} ; \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{K}(\mathbf{X}_*, \mathbf{X}_*) & \mathbf{K}(\mathbf{X}_*, \mathbf{X}) \\ \mathbf{K}(\mathbf{X}, \mathbf{X}_*) & \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I} \end{pmatrix} \right)$$

then the conditional distribution of $\mathbf{x}$ given $\mathbf{y}$ is given by

$$p(\mathbf{x} \mid \mathbf{y}) = \mathcal{N} \left(\mathbf{x}; \boldsymbol{\mu}_x + \mathbf{P}_x \mathbf{H}^\top \left(\mathbf{H} \mathbf{P}_x \mathbf{H}^\top + \mathbf{R} \right)^{-1} (\mathbf{y} - \mathbf{H} \boldsymbol{\mu}_x), \right. \\ \left. \mathbf{P}_x - \mathbf{P}_x \mathbf{H}^\top \left(\mathbf{H} \mathbf{P}_x \mathbf{H}^\top + \mathbf{R} \right)^{-1} \mathbf{H} \mathbf{P}_x \right)$$

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$$p(\mathbf{f}_\star, \mathbf{y} \mid \mathbf{X}, \mathbf{X}_\star) = \mathcal{N} \left(\begin{pmatrix} \mathbf{f}_\star \\ \mathbf{y} \end{pmatrix} ; \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) & \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) \\ \mathbf{K}(\mathbf{X}, \mathbf{X}_\star) & \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I} \end{pmatrix} \right)$$

then the conditional distribution of $\mathbf{x}$ given $\mathbf{y}$ is given by

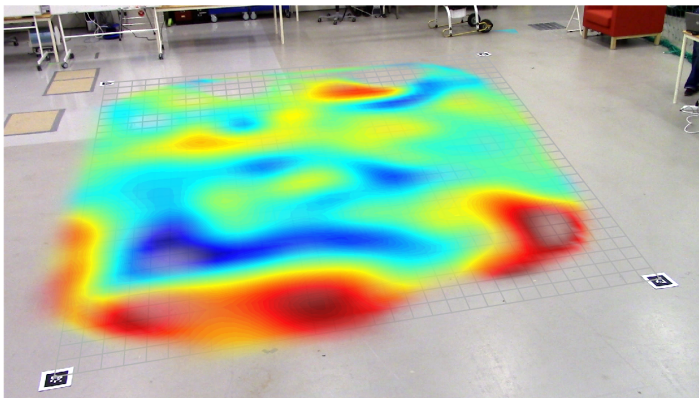
$$p(\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star) = \mathcal{N} \left(\mathbf{f}_\star ; \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y}, \right. \\ \left. \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) - \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{K}(\mathbf{X}, \mathbf{X}_\star) \right)$$

Prediction using GPs

In conclusion, we can combine a GP prior $f(\mathbf{x}) \sim \mathcal{GP}(0, \kappa(\mathbf{x}, \mathbf{x}'))$ with measurement(s) $y = f(\mathbf{x}) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$ at locations $\mathbf{X}$, to obtain predictions at previously unseen locations $\mathbf{X}_\star$ as

$$\begin{aligned}\mathbb{E} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] &= \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I}]^{-1} \mathbf{y} \\ \text{cov} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] &= \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) - \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I}]^{-1} \mathbf{K}(\mathbf{X}, \mathbf{X}_\star)\end{aligned}$$

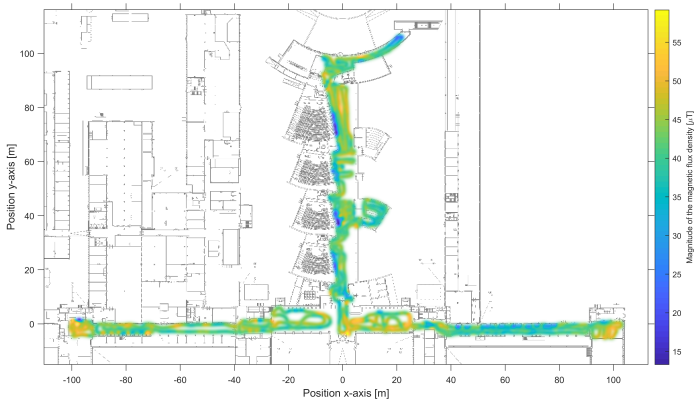
Building magnetic field maps using Gaussian process regression



<https://www.youtube.com/watch?v=enlMiUqPVJo>

Arno Solin, Manon Kok, Niklas Wahlström, Thomas B. Schön and Simo Särkkä, Modeling and interpolation of the ambient magnetic field by Gaussian processes. IEEE Transactions on Robotics, Vol. 34, Issue 4, pp. 1112-1127, 2018.

Gaussian process map of magnetic field



Piet van Beek, Marnix Fetter, Bart de Jong, Giel van der Weerd
(BEP project spring 2019)

Hyperparameters

GP priors have **hyperparameters**. For instance, for the squared exponential

$$\kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right),$$

σ_f and ℓ are hyperparameters. The measurement noise variance σ_ϵ^2 is typically also considered to be a hyperparameter.

<https://smlbook.org/GP/>

Learning hyperparameters

Assume we have a set of measurements $\mathbf{y} \in \mathbb{R}^n$ at locations $\mathbf{X}$ as $\mathbf{y} = f(\mathbf{X}) + \epsilon$, with a prior $f(\mathbf{x}) \sim \mathcal{GP}(0, \kappa(\mathbf{x}, \mathbf{x}'))$.

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$$p(\mathbf{y} \mid \mathbf{X}) = \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_{\epsilon}^2 \mathbf{I})$$

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$$p(\mathbf{y} | \mathbf{X}) = \mathcal{N}(\mathbf{0}, \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})$$

In practice we maximize the log marginal likelihood, given by

$$\begin{aligned} \log p(\mathbf{y} | \mathbf{X}) = & -\frac{1}{2} \mathbf{y}^\top (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log \det (\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I}) \\ & - \frac{n}{2} \log(2\pi) \end{aligned}$$

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Note that this is a very nonlinear optimization problem that has multiple local minima!

Different kernels

`https://smlbook.org/GP/`

Possible extensions

For standard Gaussian process regression,

- the location at which measurements are made is assumed to be known,
- the measurement noise is assumed to be Gaussian and each measurement is assumed to have the same measurement noise variance,
- the hyperparameters are learned in advance and kept constant.

But extensions exist for each of these.

Possible extensions

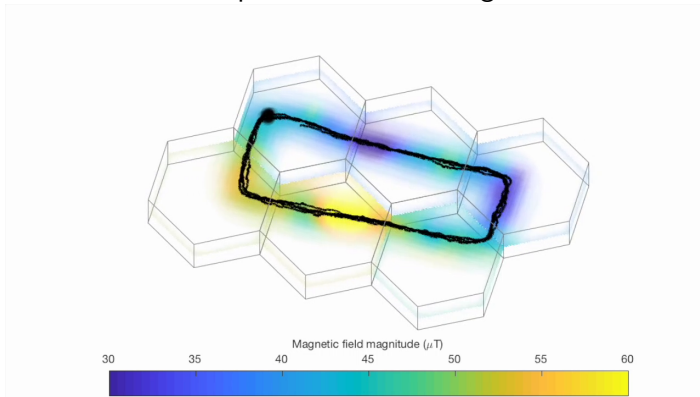
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But extensions exist for each of these.

Location at which the measurements are made unknown

Rao-Blackwellised particle filter for magnetic field SLAM



<https://youtu.be/pbwWLoh6mvI>

Manon Kok and Arno Solin, Scalable Magnetic Field SLAM in 3D Using Gaussian Process Maps. Proceedings of the 21st International Conference on Information Fusion, pp. 1353 - 1360, Cambridge, UK, July 2018.

Main challenge with GPs

Biggest challenge: Scales cubically with the number of data points!

Main challenge with GPs

Biggest challenge: Scales cubically with the number of data points!

$$\mathbb{E} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] = \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I}]^{-1} \mathbf{y}$$

$$\text{cov} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] = \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) - \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathbf{I}]^{-1} \mathbf{K}(\mathbf{X}, \mathbf{X}_\star)$$

$\mathbf{K}(\mathbf{X}, \mathbf{X})$ is a matrix with size equal to the number of measurements that we have collected.

Main challenge with GPs

Biggest challenge: Scales cubically with the number of data points!

$$\begin{aligned}\mathbb{E} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] &= \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathcal{I}]^{-1} \mathbf{y} \\ \text{cov} [\mathbf{f}_\star \mid \mathbf{y}, \mathbf{X}, \mathbf{X}_\star] &= \mathbf{K}(\mathbf{X}_\star, \mathbf{X}_\star) - \mathbf{K}(\mathbf{X}_\star, \mathbf{X}) [\mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_\epsilon^2 \mathcal{I}]^{-1} \mathbf{K}(\mathbf{X}, \mathbf{X}_\star)\end{aligned}$$

$\mathbf{K}(\mathbf{X}, \mathbf{X})$ is a matrix with size equal to the number of measurements that we have collected.

Also: E.g. $\kappa_{\text{SE}}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\frac{\|\mathbf{x}-\mathbf{x}'\|_2^2}{2\ell^2})$ is never completely zero so $\mathbf{K}(\mathbf{X}, \mathbf{X})$ will typically not be sparse.

Gaussian processes

- Multivariate Gaussian distributions
- Gaussian process priors
- Gaussian process regression